

A fundamental question

- Suppose that $v_1, \dots, v_m$ and b are vectors in R^n . How can we determine if b can be written as a linear combination of $v_1, \dots, v_m$?
- The outcome from last time is that if we form the matrix

$$A = (v_1 \ v_2 \ \dots \ v_m \ b)$$

i.e. the matrix with columns $v_1, \dots, v_m, b$, as the augmented matrix for a system of linear equations then b is a linear combination of $v_1, \dots, v_m$ iff this system is consistent.

$$v_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix} \in \mathbb{R}^n, \quad v_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{pmatrix} \in \mathbb{R}^n$$

...

$$v_m = \begin{pmatrix} a_{1m} \\ \vdots \\ a_{nm} \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in \mathbb{R}^n$$

Are there $\lambda_1, \dots, \lambda_m$ s.t.

$$\lambda_1 v_1 + \dots + \lambda_m v_m = b \quad ?$$

The λ_i 's
are the
unknown.

The augmented matrix for this lin. sys. looks like:

$$\left(\begin{array}{cccc} a_{11} & a_{12} & & a_{1m} & b_1 \\ a_{21} & a_{22} & & & \vdots \\ & & \dots & & \vdots \\ & & & & \vdots \\ a_{n1} & a_{n2} & & a_{nm} & b_n \end{array} \right)$$

$\uparrow \quad \uparrow \quad \quad \uparrow \quad \uparrow$

$\underline{v_1} \quad \underline{v_2} \quad \quad \underline{v_m} \quad \underline{b}$

$$A = (v_1 \dots v_m \ b).$$

We say that b is in the span
of $v_1, \dots, v_m$ if b is a linear comb.
of $v_1, \dots, v_m$.

$\text{Span}(v_1, \dots, v_m)$ is the set of all lin.
comb. of $v_1, \dots, v_m$.

Ex: Is $\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \leftarrow b$ a lin comb. of $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$
 $\downarrow$
 v_1

and $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} ?$
 $\downarrow$
 v_2

Is b in $\text{Span}(v_1, v_2)$?

Form $A = \begin{pmatrix} 1 & 3 & 2 \\ 0 & -1 & 2 \\ -1 & 2 & 3 \end{pmatrix}$

row reduce
 $\xrightarrow{A}$

$$\begin{pmatrix} 1 & 3 & 2 \\ 0 & -1 & 2 \\ \underline{0 & 0 & 1} \end{pmatrix}$$

← there are no solutions
to the assoc. lin.
system

So $\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ is not in the span of v_1 and v_2 .

Ex: Use the same v_1 and v_2 and let
 $b = \begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$. If $A = \begin{pmatrix} 1 & 3 & 5 \\ 0 & -1 & -1 \\ \underline{-1} & 2 & 0 \end{pmatrix}$ row
red.
→

In fact $b = 2v_1 + v_2$

Matrix equations

Suppose $v_1, \dots, v_m$ are vectors in R^n . Let that $A = (v_1 \ v_2 \ \dots \ v_m)$. Notice that A is an $n \times m$ matrix. Suppose that x is a vector in R^m say

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

Then we define

$$Ax = x_1 v_1 + \dots + x_m v_m.$$

Ax is the product of an $n \times m$ matrix with a vector in R^m and gives us an element of R^n .

Ex:

$$\begin{pmatrix} 2 & 3 \\ 1 & -1 \\ 0 & 4 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$\leftarrow \in \mathbb{R}^2$

$=$

$$\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$$

$\in \mathbb{R}^3$

A , 3×2 matrix. +

$\neq$

$$(-1) \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + (1) \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}$$

Ex:

$$3 \times 4 \rightarrow \begin{pmatrix} -1 & 3 & 0 & 4 \\ 2 & 2 & 1 & -2 \\ 1 & 1 & 2 & 1 \\ - & - & - & - \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \stackrel{\downarrow \in \mathbb{R}^4}{=} \begin{pmatrix} 21 \\ 1 \\ 13 \end{pmatrix} \stackrel{\downarrow \in \mathbb{R}^3}{=}$$

$$\begin{aligned} \text{Ex: } x + 2y + 3z &= 4 \\ -x + 3y + z &= 2 \\ 2x - y + 2z &= 1 \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ -1 & 3 & 1 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ \frac{4}{2} \\ 1 \end{pmatrix}$$

$$\begin{array}{ccc} \uparrow & \uparrow & \\ \underline{A} & \underline{x} & = \underline{b}. \end{array}$$

Restating our fundamental question

Suppose $v_1, \dots, v_m$ and b are in R^n . Let

$$A = (v_1 \dots v_m).$$

We can ask if the equation

$$\underline{Ax = b}$$

has a solution i.e. is there an x in R^m which makes this equation true. Said another way, we are asking if the associated system of linear equations has a solution.

Notice that this is the same as asking if b is in the span of $v_1, \dots, v_m$.

Suppose that we have $v_1, \dots, v_m$ in R^n . How can we tell if they span all of R^n ?

Answer: If we form $A = (v_1 \ v_2 \ \dots \ v_m)$ then $v_1, \dots, v_m$ spans R^n iff when you row reduce the matrix A , every row has a leading 1.

Important fact: If $v_1, \dots, v_m$ spans R^n then $m \geq n$.

Properties of Multiplication of Matrices and Vectors

Theorem

Suppose that A is $n \times m$ and x, y are in R^m .

- 1 $A(x + y) = Ax + Ay$ and,
- 2 for every number λ , $A(\lambda x) = \lambda Ax$.